



Pool boiling heat transfer enhancement with copper nanowire arrays



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HIGHLIGHTS

- We investigated the effect of nanowire arrays on pool boiling heat transfer enhancement.
- Nanowires with five different lengths were electroplated on bare copper surface.
- We studied the influence of the length of copper nanowire arrays on HTC and CHF of boiling.
- We explored the physics of the boiling mechanism on nanostructure surfaces.

ARTICLE INFO

Article history:

Received 11 January 2014

Accepted 15 September 2014

Available online 20 September 2014

Keywords:

Boiling heat transfer

Cu nanowire array

HTC

CHF

ABSTRACT

The pool boiling heat transfer on copper nanowire arrays has been experimentally studied. Five different copper nanowire arrays with various lengths (3 μm , 5 μm , 10 μm , 20 μm , and 30 μm) have been electroplated on bare smooth copper surfaces. Their pool boiling heat transfer performances were measured and compared with control surfaces. It is found that the integration of copper nanowire arrays on heating surface can effectively enhance the boiling heat transfer coefficient (HTC) as well as the critical heat flux (CHF). As the length of nanowires increases, more enhancements have been observed. This augment is due to the enhancement of the wettability and the number of the nucleation sites (defects).

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1. Introduction

Boiling is the most efficient mode of heat transfer due to its inherent phase change nature. Hence, it is considered as one of the solutions capable for cooling electronic devices with extra-high heat flux in future.

Boiling heat transfer enhancement has been researched extensively over the last century. A detailed review has been provided by Pioro et al. [1]. One approach to enhance boiling heat transfer is to manipulate the topography of boiling surface. Several commercial products have proved that artificial surfaces, such as porous sintered coating (Union Carbide HIGH-FLUX), re-entrant grooves (Wielanderke AG GEWA-T), and tunneled surfaces formed by bending notched fins (Hitachi, THERMOEXCEL-E), can effectively raise the boiling Heat Transfer Coefficient (HTC) and the Critical Heat Flux (CHF) comparing with untreated bare smooth surfaces. This traditional enhancement was attributed to the increase of the

liquid–solid contact area and the number of the nucleation sites in size of 10 μm .

Recently, researchers have noticed that the nanostructures can also improve the pool boiling heat transfer performance. In 2006, Ahn et al. [2] reported that the boiling surface integrated with vertically aligned multiwall carbon nanotubes (CNTs) can have an augment of CHF by 25–28% for PF-5060, compared with the control surfaces. Ujereh et al. [3] synthesized CNTs on copper and silicon surfaces and measured the pool boiling heat transfer performance in FC-72 experimentally. They found that the fully CNTs coated surfaces showed a noticeable decrease in the incipience superheat as well as a shift of the entire nucleation boiling region (about 2 °C) toward lower wall superheats. The similar conclusion has also been reported in Sathyamurthi et al.'s researches [4] for sub-cooled pool boiling. In their experiments, it was noticed that MWCNTs coated substrates yields higher wall heat fluxes under saturated and sub-cooled conditions compared with a bare silicon surface, while the enhancement heat flux was weakly dependent on the thickness of the MWCNTs. In 2009, Khanikar et al. [5] applied CNTs in micro-channels and the similar results had been observed.

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These researches provide an approach to enhance the boiling heat transfer performance by introducing CNTs on heating surfaces. However, the synthesis of CNTs on desired surfaces by chemical vapor deposition (CVD) requires high temperature environment (usually 500 °C–700 °C), which is not capable for applications in electronic devices. Recently, researchers found that other nanowires, such as copper, TiO_2 , silicon, had similar effects on boiling heat transfer performance while the synthesis can be achieved in low temperatures. In 2008, Li et al. [6] deposited a copper nanowire layer on a substrate using electron-beam evaporation method. The pool boiling experiments showed that the deposition of copper nanorods on polished copper substrate brought more than 30% enhancement to the HTC. However, an apparent CHF rise was not observed in their experiments. While in Chen et al.'s work [7], both the HTC and the CHF enhancements had been reported by introducing copper nanowires as well as silicon nanowires on smooth silicon substrate. It is noticeable that their copper nanowires were synthesized by electroplating, which can be conducted at room temperature. Yao et al. [8] developed a new technique to directly grow Cu/Si nanowires with various heights. They obtained a heat flux of $134\text{W}/\text{cm}^2$, which was 300% higher than a plain Si surface at the same wall superheat. Some other observations, such as Chen et al.'s [9], Im et al.'s [10] and Thiagarajan et al.'s [11], have also been published in last several years. Different nanowires have been integrated on boiling surfaces. Most of them have found that both the HTC and CHF had been improved by introducing nanowire arrays on bare smooth surfaces.

Though various experiments have proved the significant augment of boiling heat transfer performance using nanowire arrays, the mechanisms of such enhancement are still in debate. Several arguments have been proposed: 1) The wettability of the treated surface has been effectively improved. The contact angle decreases, which causes an increased CHF through the enhanced liquid spreading over the heated area. In the meantime, it also

expands the contact area between solid and liquid. The fin effect reduces the interface temperature and hence shifts the boiling curve to the left side. 2) The abundant micro-sized cavities in the nanowire arrays defects can effectively trap air or vapor and serve as nucleation sites, which reduce the wall superheat. They also can provide sites for stable vapor formation at the top of the nanowire coating, which alters the critical distance between vapor columns on the heating surface and thus adjusts the critical instability wavelength. 3) The introducing of nanowires enhances the capillary pumping effect on solid surfaces, thereby delaying the dry out of the region under the vapor column.

According to these impacts, the effect of the length of nanowire arrays must affect the boiling heat transfer performance dramatically since some critical facts, such as wettability, number of defects and capillary pumping ability, all depend on the length. However, such researches have barely conducted and the pool boiling performance of nanowire array surfaces on the nanowire length is not revealed yet. This paper focuses on the preliminary experimental studies of the influence of the length of copper nanowire arrays on HTC and CHF of boiling and explores the physics of the boiling mechanism on nanostructure surfaces.

2. Experiment

2.1. Experimental setup

Fig. 1 shows the apparatus for the pool boiling heat transfer experiments. A copper heat transfer block ($1\text{ cm} \times 1\text{ cm}$ in cross section) embedded with a heater unit at the lower section was built and submerged in the test fluid (DI water), which was then heated up to its saturation temperature (100 °C) at 1 atm. The top surface, which was synthesized with copper nanowire array on, was directly contacted with test fluid, while the other surfaces were protected with ceramic fiber insulation materials from heat loss.

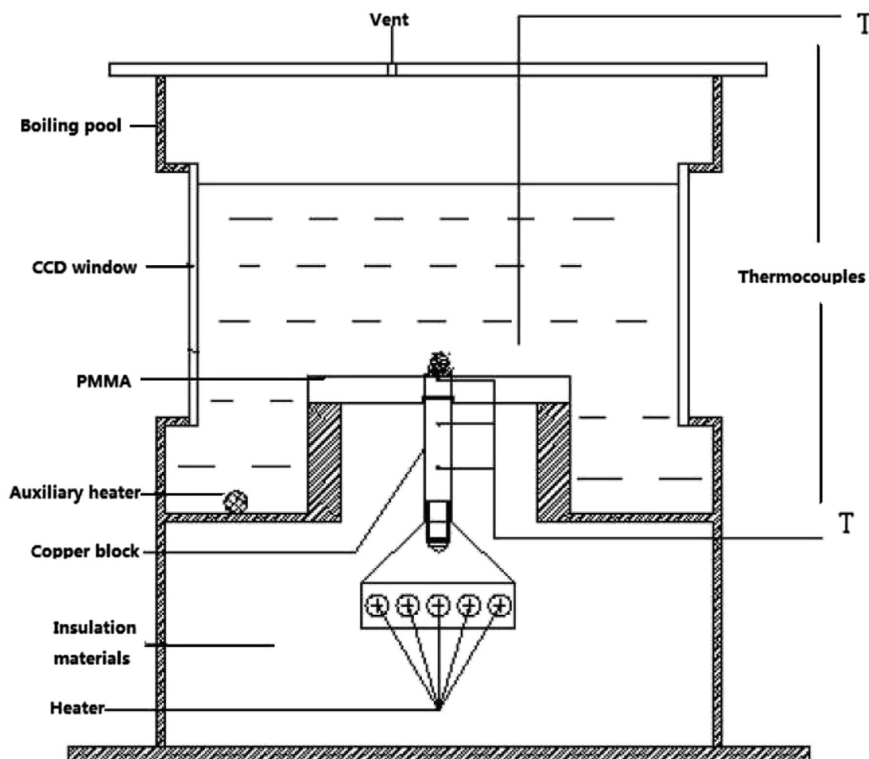


Fig. 1. Schematic of the experimental set-up.

The heater unit composed 5 cartridge heaters, which were placed on the bottom of the copper block substrate and could supply heat up to 750 W. A variable transformer was used to control the heat flux by regulating the input voltage during the experiments. Several k-type thermocouples with a diameter of 0.5 mm were embedded inside the copper block (centers of cross section) to measure the axial heat flux. The surface temperature T_s , which is in the interface of copper block and test liquid underneath the copper nanowire array, can be deduced using linear extrapolation of the temperatures measured by these thermocouples (Fig. 2). The temperature of the test fluid, T_{sat} , was monitored by a thermocouple located about 1 cm from the boiling surface in the pool. With T_s and T_{sat} , the wall superheat can be deduced using $\Delta T = T_s - T_{\text{sat}}$. In order to observe the status of bubbles on the boiling surface, a size of 100 mm $\times$ 100 mm CCD window was installed on each side of the stainless steel container. A vent was opened on the top of the boiling pool to guarantee the experiment conducted under atmospheric pressure. The whole boiling vessel was heated by an external auxiliary heater attached surround to assure that the bulk test fluid was kept saturated throughout the whole experiments.

The heat transfer coefficient will increase dramatically as the liquid level drops if DI water level in the boiling pool is lower than the critical level [12]. In this experiment, the critical level was examined and determined to be 5 mm after several tests. During the whole test, part of the steam condensed on the inner top surface of the boiling pool and returned to the container. In the meantime, a certain amount of saturated boiling DI water was added into the container regularly to make sure that the water level was always much higher than the critical level.

2.2. Sample preparation

In order to reduce the impact of thermal contact resistance, Cu nanowire arrays are synthesized directly on the top surface of the copper block via electroplating using a commercial available porous alumina membrane (PAA) template (Whatman Inc.), as shown in Fig. 3.

The Electroplating process in this study includes two steps. Before synthesizing, the top surface of copper block was well polished by 300#, 600#, 1000#, 1400# and 2000# sand paper in a sequence. Then it was cleaned using acetone and acid to further remove contamination and oxidation. A PAA template, filter paper and the counter electrode of copper were placed in an order and fixed by clamps, which is shown in Fig. 4(a). The edge of the filter paper was immersed in the plating solution to ensure that the solution could be absorbed fully by the filter paper through capillarity and distributed uniformly in the PAA template. In the present

study, the PAA template is 47 mm in diameter and 50 μm thick with 200 nm pores. The porosity of the PAA template is about 50%. A constant potential is applied between the copper substrate and the counter electrode to electroplate nanowires. After a certain time, one layer of short copper nanowire arrays is attached on the copper substrate.

The filter paper and the clamps were removed and the others were submerged into the copper pyrophosphate plating solution to electroplate in the second step, as shown in Fig. 4(b). The solution contains $\text{Cu}_2\text{P}_2\text{O}_7$ as the provider of the main salt for copper ions which can affect the uniformity of the coating, $\text{K}_4\text{P}_2\text{O}_7$ as the main bath complexing agent which can improve the conductivity of the plating solution and $\text{C}_6\text{H}_5\text{O}_7(\text{NH}_4)_3$ as the auxiliary complexing agent which can improve the dispersibility of the plating solution.

A three-electrode system is adopted to obtain nanowires of different lengths in the second step since it can control the potential more precisely. The substrate of copper is treated as the working electrode and the counter electrode is served as the auxiliary electrode, while Ag/AgCl electrode is selected as the reference electrode. The above three electrodes are connected to an electrochemical analyzer (CHI608D, CH Instruments) which is controlled by a computer. The electroplating time and the voltage applied between the working electrode and the auxiliary electrode is adjusted to corresponding value to achieve desired nanowire lengths.

The temperature of the plating solution was maintained at room temperature through a water bath. A stirrer was applied to guarantee a uniform ions distribution in the plating solution throughout the whole experiment.

In this work, five different nanowire arrays with various lengths (3 μm , 5 μm , 10 μm , 20 μm , and 30 μm) had been fabricated. After the Cu nanowire array was deposited, the templates were removed by wet etching in NaOH solution to obtain free standing Cu nanowire array (Fig. 5(b), (c), and (d)). The copper blocks were then rinsed in DI water, dried in a vacuum chamber and installed to the test facility.

2.3. Experimental procedure

Pool boiling experiments were conducted under steady-state conditions. The voltage input, which was controlled by the variable transformer, was adjusted from 0V to higher voltage in an increment of 3V to obtain the desired heating power. For each heating power, the system was kept for about half an hour waiting time to make sure that the steady-state condition was satisfied. The measured data of thermocouples were recorded with a data acquisition (Keithley2700) and reduced by a computer. The Critical Heat Flux was postulated to be equal to the heat flux corresponding to the last observed stable temperature measurement, beyond which a sudden dramatic rise of the temperature of the copper block can be monitored.

3. Results and discussion

3.1. Boiling performance of various surface geometries

To eliminate the affection of the residual defects on heating surface, a reference (control) experiment was performed at first using polished bare smooth copper surface (same polish procedure as described above). As suggested by many researchers [9,11,13–22], different surface geometries result in diverse boiling performances. The control experiment therefore provides a consistent reference for heat transfer augment due to presence of the copper nanowire arrays.

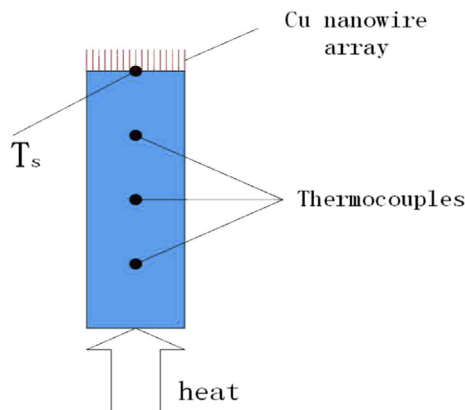


Fig. 2. Heat flux and surface temperature measurements.

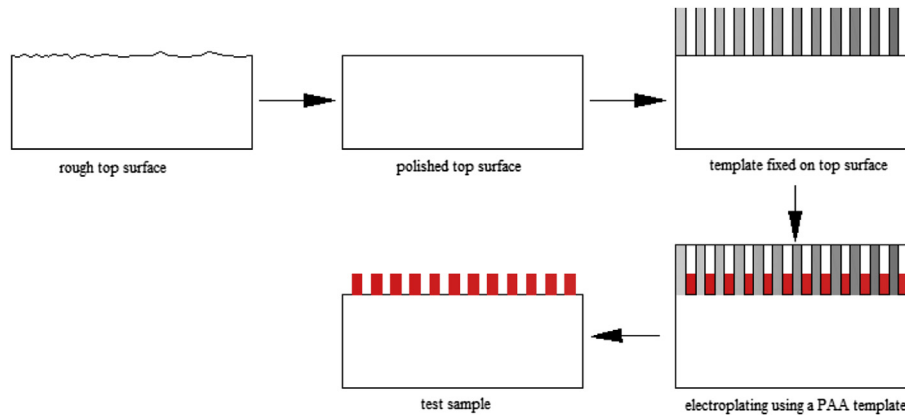


Fig. 3. The synthesis process of copper nanowire array on the top surface of copper block.

Experiments were conducted for five surfaces integrated with various lengths of nanowire arrays (3 μm , 5 μm , 10 μm , 20 μm , 30 μm). The results are depicted in Fig. 6. Boiling on plain smooth copper surface serves as the primary control for the boiling performance comparison. It can be seen that the introducing of the nanowire arrays shifts the boiling curve to the top left corner, which means both the HTC and the CHF have been significantly enhanced. As shown in Fig. 6, it is also obvious that the length of nanowire arrays has positive effects on boiling heat transfer. The improvement of CHF along with the increase of nanowire lengths is obvious, while the larger gradient implies the HTC enhancement, as in Fig. 6. Both the HTC and CHF increase with the length of nanowires.

It is also can be seen from Fig. 6 that under the same wall superheat, heat flux on surfaces of copper nanowire arrays has a significant increase compared to the smooth surface. For instance, when the wall superheat is 15 $^{\circ}\text{C}$, the critical heat flux on the nanowire (30 μm) surface is up to 140W/cm², which is more than three times of that of the smooth surface. The copper nanowire

arrays can also reduce the wall superheat under a certain heat flux, for example, when the surface heat flux is 100W/cm², the superheat of smooth surface is about 20 $^{\circ}\text{C}$, which is higher than that of copper nanowire (30 μm) surfaces by 7 $^{\circ}\text{C}$. In addition, the incipient wall superheats of copper nanowire surfaces are also lower than that of the smooth surface, which further illustrates the outstanding heat transfer performance of copper nanowire surfaces. This result is in agreement with Yao et al.'s [8]. Some other similar enhancements have also been observed in Refs. [7,23,24].

3.2. Thermal performance analysis

This trend may be attributed by two main impacts.

3.2.1. The impact of nanowires on nucleation sites

Firstly, it is noted that the distances between adjacent nanowires are only in hundreds of nanometers, which are much smaller than the effective size (several micrometers) for nucleation, so these spaces are hard to become vaporization cores. However, unlike the idealized smooth surfaces, most real solid surfaces, as shown in Fig. 7(a) and (b), inherently contain natural or machine-formed pits, scratches, or other irregularities, the size of which may range from microscopic to macroscopic. Some of these imperfections are in proper geometry and are in tens of micrometers size as depicted in Fig. 7(a) that are quite difficult for liquid to permeate into. It is expected that such cavities can contain entrapped gas and vaporization may occur at the liquid–gas interface in these cavities at relatively low temperatures [25]. Entrained gas in cavities serves to provide nuclei for the formation of vapor bubbles at the onset of vaporization. Thus it is quite possible that these defects may turn into effective nucleation sites. For some cavities as shown in Fig. 7(b), the cavity mouth angle Θ is larger than α , where α is the contact angle of water on bare smooth copper surface. These cavities can easily be filled with liquid, leading to less vaporization sites. However, as illustrated in Fig. 7(c), some of the defects can be protected by nanowires electroplated nearby from being filled with liquid and thus turn into effective vaporization cores due to the decrease of the cavity mouth angle Θ . It can be seen from Fig. 7(c), when the nanowires are short (5 μm), the alignment shows in a good order. However, if the nanowires get longer, one can see that the nanowires collapse, as shown in Fig. 5(d) and Fig. 7(d). Some defects with large open mouths may survive and become effective vaporization cores, resulting in more and larger nucleation sites. Due to the collapse, numerous and widely distributed cavities with sizes of several micrometers appear in between (Fig. 5(c) 10 μm , Fig. 5(d) 20 μm , Fig. 7(d) 20 μm ,

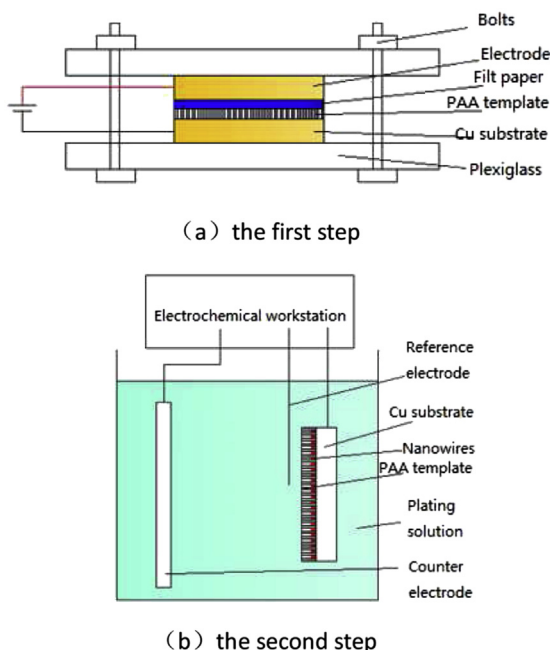


Fig. 4. Schematic of fabricating copper nanowires.

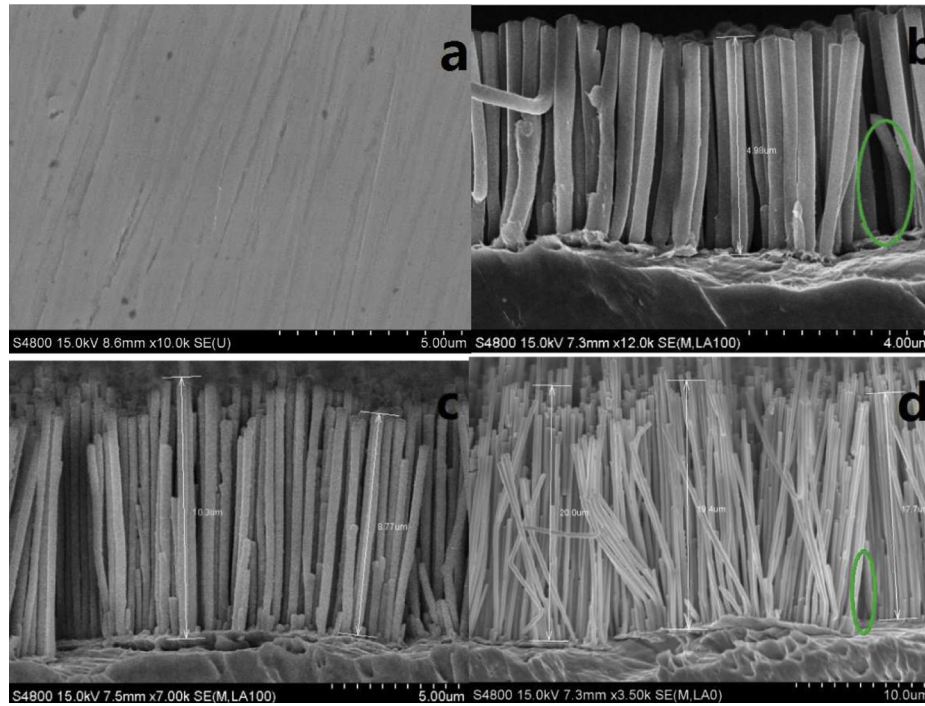


Fig. 5. SEM photograph of a) smooth surface (top view), b) 5 μm NWs, c) 10 μm NWs, and d) 20 μm NWs.

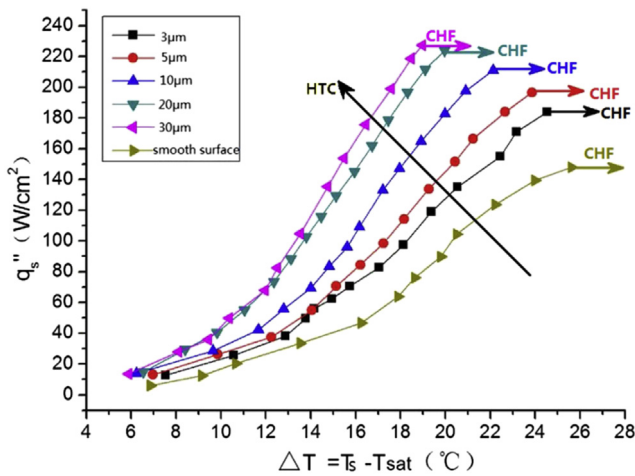


Fig. 6. Boiling curves of the reference Cu Plate and of surfaces with nanowire arrays.

Fig. 8(a) 5 μm and Fig. 8(b) 10 μm), bringing in more vaporization cores.

3.2.2. The impact of nanowires on surface wettability

Several researchers [7,11,26–28] have measured the contact angle on the surface of nanowire arrays before. In this study, the surface contact angles of DI water on copper nanowire arrays have

also been measured. The results indicate that surfaces with longer nanowires are more hydrophilic than those with shorter ones, as shown in Fig. 9. When the length of the copper nanowires is 3 μm , the contact angle (α_a) is approximately 29° . Whereas, when the length becomes 30 μm , the contact angle (α_b) shrinks to 19° . According to the theoretical model proposed by Dhiri and Liaw [29], the HTC on the boiling surface increases as the surface contact angle decreases. In this article, the surface contact angle decreases as the lengths of copper nanowires increase. Hence the HTC on the copper nanowire surfaces increases as the nanowire length increases. The hydrophilic surface reduces bubbles attaching time [30], which increases the bubble frequency and enhances heat transfer. This is reasonable as heat transfer in boiling is dominated by bubble dynamics [29] rather than heat conduction. In the experiments, the visual observations proved this argument.

Furthermore, because of the hydrophilic property, the evaporation micro-layer of bubbles spreads larger, resulting in a smaller dry area and higher evaporation efficiency [31]. In addition, the nanowire shows a kind of thermal fin effect, which further improves the heat transfer performance.

4. Conclusions

The pool boiling heat transfer on copper nanowire arrays have been experimentally studied. A reference (control) experiment was performed at first using polished bare smooth copper surface. Five different copper nanowire arrays with various lengths, 3 μm , 5 μm ,

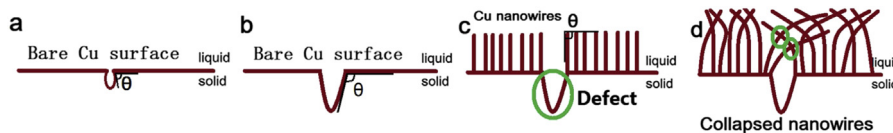


Fig. 7. Schematic representation of vapor and liquid entrapment of different surface geometries a) bare Cu surface with effective defects b) bare Cu surface with invalid defects c) Surface with short Cu nanowires (5 μm) d) Surface with long Cu nanowires (20 μm).

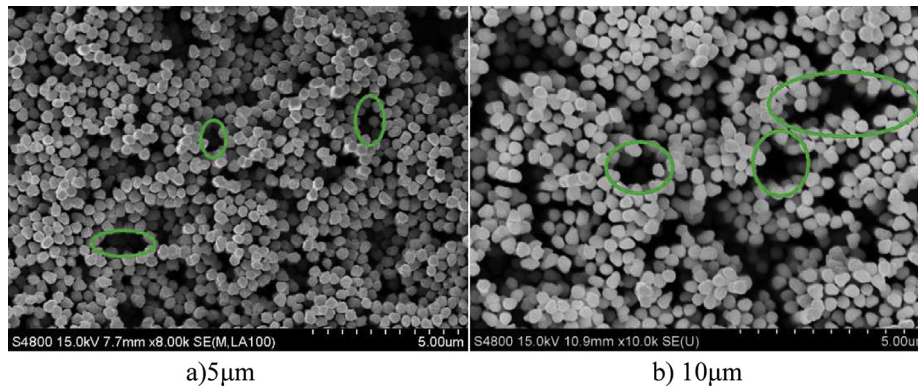


Fig. 8. Defects in nanowire arrays (top view, SEM graph).

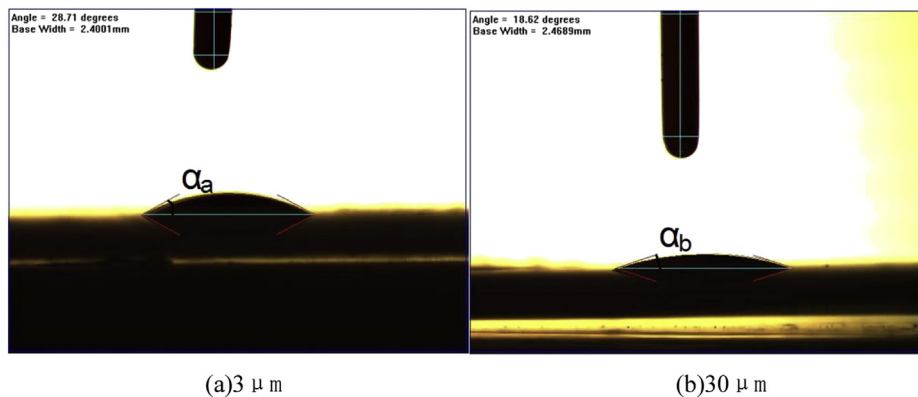


Fig. 9. Contact angle of water droplets on nanowire arrays.

10 μm, 20 μm, and 30 μm, have been deposited on bare smooth copper surfaces by electroplating. Their pool boiling heat transfer performances were measured and compared with control surfaces. It is found that the heating surfaces with copper nanowire arrays have higher boiling heat transfer coefficient as well as critical heat flux. As the length of nanowires increase, more enhancements have been observed. This augment is due to the increase of the surface wettability and the number of the nucleation sites (defects) introduced by collapsing of nanowires.

Acknowledgements

This work was supported by New Teachers' Fund for Doctor Stations, Ministry of Education of China (20103219120031).

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